AIR QUALITY INDEX FORECASTING USING

GA-KELM

A MINI PROJECT REPORT

Submitted

in the partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE AND ENGINEERING

by

D. Akhila (23B81A0502)

K. Jagruthi (23B81A0516)

K. Yashashwini (23B81A0563)

Under the Guidance of

N.Yadagiri

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CVR COLLEGE OF ENGINEERING

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Vastunagar, Mangalpalli (V), Ibrahimpatnam (M),

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the Industrial Oriented Mini project entitled "**AIR QUALITY INDEX FORECASTING USING GA-KELM**" being submitted by **D.Akhila (23B81A0502), K.Jagruthi (23B81A0516), K.Yashashwini (23B81A0563)** in partial fulfillment for the award of Bachelor of Technology in Computer Science and Engineering to the CVR College of Engineering, is a record of bona fide work carried out by them under my guidance and supervision during the year 2025-2026.

The results embodied in this Industrial Oriented Mini project work have not been submitted to any other University or Institute for the award of any degree or diploma.

Signature of the project guide	Signature of the HOD
N.Yadagiri	**Dr. M.Raghava**
Assistant Professor	**Department of CSE**

External Examiner

DECLARATION

I hereby declare that this project report titled **"Air Quality Index Forecasting using GA-KELM"** submitted to the Department of Computer Science and Engineering, CVR College of Engineering, is a record of original work carried out by me. The information and data presented in this report are true and authentic to the best of my knowledge. This project has not been submitted to any other university or institution for the award of any degree or diploma, nor has it been published at any time before.

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D.Akhila (23B81A0502)
K. Jagruthi (23B81A0516)
K. Yashashwini (23B81A0563)

Date:
Place:

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ABSTRACT

Air quality has become one of the most critical environmental and public health concerns across the globe. Rapid industrialization, urbanization, and increased vehicular emissions have significantly contributed to deteriorating air quality levels, making it essential to develop accurate prediction systems. Traditional methods for forecasting Air Quality Index (AQI) often rely on statistical models or standalone machine learning approaches, which fail to capture the complex nonlinear relationships between pollutants and environmental factors.

This project introduces a hybrid model known as **Genetic Algorithm-based Kernel Extreme Learning Machine (GA-KELM)** for efficient and accurate AQI prediction. The proposed system combines the fast learning capability of Extreme Learning Machine (ELM), the nonlinear mapping strength of Kernel methods, and the optimization power of Genetic Algorithms (GA). The model is capable of predicting the concentrations of six major pollutants: SO_2, NO_2, PM10, PM2.5, O_3, and CO.

Unlike traditional models, GA-KELM optimizes both the structure and parameters of the learning model, reducing prediction errors and improving stability. The results demonstrate that the proposed approach achieves faster training and higher prediction accuracy compared to existing models such as CMAQ, SVR, and DBN-BP.

Furthermore, the GA-KELM model demonstrates strong generalization ability across different datasets and seasonal variations, making it suitable for real-world applications. The system is designed to provide reliable and consistent predictions, which can assist government authorities, environmental agencies, and the public in taking preventive measures to reduce air pollution exposure.

Overall, this project presents an intelligent, efficient, and scalable solution for AQI forecasting, contributing toward better environmental monitoring and decision-making for sustainable urban development.

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CHAPTER 1
INTRODUCTION

Air pollution is a growing environmental issue that directly affects human health, ecosystems, and climate conditions. According to global studies, air pollution is responsible for millions of premature deaths each year due to respiratory and cardiovascular diseases. The rapid increase in industrialization, urbanization, and vehicular emissions has significantly contributed to the degradation of air quality across both developed and developing countries. In addition to human health impacts, air pollution also affects agricultural productivity, biodiversity, and climate patterns. Monitoring and predicting air quality has therefore become essential for implementing preventive measures, raising public awareness, and supporting government policies aimed at pollution control.

Air Quality Index (AQI) is a standardized metric used to represent the concentration levels of multiple pollutants in the air in a simplified manner. It provides an easy-to-understand indication of how polluted the air is and what associated health risks may arise for different population groups. The AQI is calculated based on the maximum value of individual pollutant indices, considering major pollutants such as SO_2, NO_2, PM10, PM2.5, O_3, and CO. These pollutants originate from various sources including industrial emissions, transportation, construction activities, and natural phenomena. Accurate AQI measurement and forecasting help individuals take precautionary measures, such as reducing outdoor activities during high pollution levels, thereby minimizing health risks.

To overcome these challenges, this project proposes a hybrid intelligent system using GA-KELM. The integration of Genetic Algorithms with Kernel Extreme Learning Machines allows the system to automatically optimize model parameters, including weights, thresholds, and kernel functions, thereby improving both prediction accuracy and computational efficiency. Additionally, the model incorporates meteorological factors and time-series data to better understand pollutant behavior and seasonal trends. This approach results in a fast, stable, and reliable prediction system that can be used for real-time AQI forecasting, supporting environmental management, policy-making, and public health protection.

1.1Motivation

The rapid increase in air pollution has become a serious environmental and public health issue worldwide. Industrial growth, urbanization, and the rising number of vehicles have significantly contributed to poor air quality, particularly in densely populated regions. Exposure to polluted air can lead to severe health problems such as respiratory diseases, cardiovascular disorders, and even premature death. This growing concern creates a strong need for effective systems that can monitor and predict air quality in advance, helping people and authorities take necessary precautions.

Another important motivation for this project is the limitations of existing AQI prediction methods. Traditional statistical models are not capable of capturing the complex and nonlinear relationships between pollutants and environmental factors. Although machine learning approaches have improved prediction accuracy, they often suffer from issues such as slow training, overfitting, and instability due to improper parameter selection. These limitations reduce their effectiveness in real-time applications and highlight the need for a more reliable and efficient prediction model.

To overcome these challenges, this project focuses on developing an advanced AQI prediction system using GA-KELM. By integrating Genetic Algorithms with Kernel Extreme Learning Machines, the model can automatically optimize its parameters and improve prediction accuracy and stability. Additionally, the system considers multiple influencing factors such as meteorological conditions, making it more adaptable to real-world scenarios. This motivates the development of a smart, efficient, and scalable solution for air quality forecasting and environmental management.

1.2 Problem Statement

Air pollution prediction is a challenging task due to the complex interactions between various environmental and meteorological factors. Existing models often struggle to maintain accuracy under changing seasonal conditions and fluctuating pollutant levels. Single machine learning models are not sufficient to handle such variability, leading to inconsistent predictions. The presence of dynamic factors such as temperature variations, humidity changes, wind patterns, and human activities further complicates the prediction process, making it difficult for conventional approaches to provide reliable results.

One of the major issues with Extreme Learning Machine (ELM) is its reliance on randomly assigned input weights and hidden layer parameters, which results in unstable outputs. Although Kernel ELM (KELM) improves prediction accuracy, it requires careful tuning of hyper parameters such as the regularization parameter (C) and kernel parameter (γ). However, no existing system effectively optimizes both discrete and continuous parameters simultaneously. This lack of proper optimization leads to reduced model performance, increased prediction errors, and limited adaptability across different datasets and environmental conditions.

Therefore, there is a need for a robust, stable, and efficient model that can automatically optimize parameters, handle nonlinear relationships, and deliver accurate AQI predictions in real-time. The GA-KELM model addresses these challenges by combining evolutionary optimization with kernel-based learning. By utilizing Genetic Algorithms, the model can intelligently search for optimal parameter values, reducing dependency on manual tuning and improving overall system performance.

1.3 Project Objectives

Develop an Efficient AQI Prediction System: Design and implement a robust system using GA-KELM to accurately predict Air Quality Index values based on pollutant and meteorological data, ensuring improved environmental monitoring.

Predict Multiple Air Pollutants: Build a model capable of predicting concentrations of major pollutants such as SO_2, NO_2, PM10, PM2.5, O_3, and CO, providing a comprehensive assessment of air quality.

Improve Prediction Accuracy: Enhance the accuracy of AQI forecasting by capturing complex nonlinear relationships between pollutants and environmental factors using kernel-based learning.**Enable Skill Gap Optimize Model Parameters Automatically:** Utilize Genetic Algorithms to automatically optimize parameters such as weights, thresholds, hidden neurons, and kernel parameters, reducing manual intervention and improving model stability.

Incorporate Meteorological Factors: Integrate environmental variables such as temperature, humidity, wind speed, and pressure into the prediction model to improve reliability and real-world applicability.

Ensure Fast Training and Computation: Develop a system that achieves faster training compared to traditional neural networks, making it suitable for real-time AQI prediction applications.

Improve Model Stability and Consistency: Reduce instability caused by random parameter selection in traditional models by applying optimization techniques, ensuring consistent prediction results across multiple runs.

Perform Data Preprocessing Effectively: Implement techniques such as missing value handling, outlier removal, and normalization to improve data quality and enhance model performance.

Evaluate Model Performance: Assess the effectiveness of the model using performance metrics such as RMSE, MSE, and R^2, and compare results with existing models like CMAQ, SVR, and DBN-BP.

Support Real-Time Environmental Monitoring: Enable the system to provide timely AQI predictions that can assist authorities and individuals in taking preventive actions against air pollution.

Promote Environmental Awareness and Decision-Making: Provide meaningful insights into air quality trends to support better environmental planning, policy-making, and public health protection.

1.4 Project Report Organization

This project report presents a comprehensive overview of the design, development, and implementation of the Air Quality Index (AQI) prediction system using the GA-KELM model. The report is structured in a systematic manner to provide a clear understanding of the problem, methodology, and results. Each chapter is organized to explain different aspects of the project, starting from basic concepts to advanced implementation details, ensuring a smooth flow of information for the reader.

Chapter 1 introduces the concept of air pollution and its impact on human health and the environment. It explains the importance of AQI prediction and outlines the motivation, problem statement, and objectives of the project. This chapter provides the necessary background and highlights the need for an efficient and accurate prediction system using advanced techniques like GA-KELM.

Chapter 2 presents the literature review, discussing various existing methods and technologies used for air quality prediction. It analyzes traditional statistical models, machine learning approaches, and deep learning techniques, along with their advantages and limitations. This chapter helps in identifying research gaps and justifies the need for the proposed GA-KELM model.

Chapter 3 describes the system requirements, including both software and hardware specifications needed to implement the project. It also explains the tools, technologies, and datasets used, along with the performance metrics for evaluating the model. This chapter ensures that all necessary resources for the project are clearly defined.

Chapter 4 focuses on the proposed system design and methodology. It explains the working of the GA-KELM model in detail, including data preprocessing, kernel-based learning, and genetic algorithm optimization. It also includes system architecture, flowcharts, and diagrams to illustrate the workflow and structure of the system.

Chapter 5 covers the implementation and experimental results of the project. It describes how the model is trained and tested using real-world datasets and presents the results obtained. This chapter also includes performance comparisons with existing models and discusses the effectiveness of the proposed approach.

Chapter 6 concludes the report by summarizing the overall work and highlighting the key contributions of the project. It also discusses possible future enhancements, such as integrating real-time data, improving model performance, and extending the system for broader environmental monitoring applications.

CHAPTER 2
LITERATURE REVIEW

2.1 Existing Work

Air Quality Index (AQI) prediction has been an active area of research due to its importance in environmental monitoring and public health protection. Over the years, several techniques have been developed ranging from traditional statistical models to advanced machine learning and deep learning approaches. Each method has contributed to improving prediction accuracy, but also comes with certain limitations when applied to real-world dynamic environments.

Initially, statistical models such as linear regression and time-series analysis were widely used for AQI prediction. These models rely on historical pollutant data to establish mathematical relationships between variables and predict future values. They are simple to implement and computationally efficient, making them suitable for basic forecasting tasks. However, air quality data is highly nonlinear and influenced by multiple dynamic factors such as weather conditions and human activities. As a result, statistical models often fail to capture complex relationships, leading to lower prediction accuracy under varying environmental conditions.

To overcome these limitations, machine learning techniques such as Artificial Neural Networks (ANN) and Support Vector Regression (SVR) were introduced. These models are capable of learning complex patterns from large datasets and provide better prediction accuracy compared to traditional methods. Machine learning approaches can effectively handle nonlinear relationships between pollutants and environmental factors. However, they still face challenges such as slow training processes, sensitivity to parameter selection, and the risk of overfitting. These issues limit their performance, especially when applied to real-time AQI prediction systems.

With further advancements, deep learning models such as Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks have been applied to AQI prediction. These models are highly effective in capturing spatial and temporal dependencies in data, allowing for more accurate forecasting of pollutant levels. LSTM models, in particular, are well-suited for time-series data and can model long-term dependencies in air quality trends. Despite their advantages, deep learning models require large volumes of data and significant computational resources, making them complex and time-consuming to implement. Additionally, their lack of interpretability makes it difficult to understand how predictions are generated.

Extreme Learning Machine (ELM) was introduced as a faster alternative to traditional neural networks. It is a single hidden layer feedforward network that significantly reduces training time by randomly assigning input weights and biases, and directly computing output weights. While ELM provides faster learning and good generalization, its reliance on random parameter initialization often results in unstable predictions and reduced accuracy.

To improve the performance of ELM, Kernel Extreme Learning Machine (KELM) was developed by incorporating kernel functions similar to those used in Support Vector Machines. KELM transforms input data into a higher-dimensional space, enabling better handling of nonlinear relationships and improving prediction accuracy. However, the performance of KELM depends heavily on the selection of kernel parameters such as the regularization parameter (C) and kernel parameter (γ). Improper selection of these parameters can lead to suboptimal results.

To address parameter optimization issues, hybrid approaches combining Genetic Algorithms (GA) with ELM-based models have been proposed. Genetic Algorithms are inspired by natural evolution and are used to search for optimal solutions through processes such as selection, crossover, and mutation. When combined with ELM or KELM, GA helps in optimizing model parameters and improving prediction performance. However, many existing hybrid models do not fully optimize all parameters or fail to balance computational efficiency and accuracy.

2.2 Limitations of Existing Work

Despite significant progress in AQI prediction techniques, existing methods still face several challenges. Statistical models lack the ability to handle complex nonlinear relationships, while machine learning models require careful parameter tuning and may suffer from instability. Deep learning approaches, although accurate, demand high computational resources and large datasets, making them less suitable for real-time applications.

Furthermore, models like ELM suffer from instability due to random parameter selection, and KELM requires proper tuning of hyperparameters for optimal performance. Existing hybrid approaches attempt to address these issues but often fail to optimize both model structure and parameters simultaneously. These limitations highlight the need for a more robust, efficient, and optimized prediction model.

To overcome these challenges, the proposed GA-KELM model integrates kernel-based learning with genetic optimization, enabling automatic parameter tuning, improved stability, and higher prediction accuracy. This approach provides a balanced solution that is both efficient and suitable for real-time AQI prediction**.**

CHAPTER 3

SOFTWARE & HARDWARE SPECIFICATION

3.1 Software Requirements

- Operating System: Windows 10/11, Linux (Ubuntu)
- Programming Language: Python (Version 3.8 or above)
- Libraries/Frameworks:
  - NumPy (for numerical computations)
  - Pandas (for data preprocessing)
  - Scikit-learn (for machine learning models)
  - Matplotlib / Seaborn (for visualization)
- Development Tools:

  Visual Studio Code
- Frontend : tilwind CSS,JavaScript
- Dataset Source: AQI datasets
- Version Control (Optional): Git
- Internet Requirement: Required for dataset access and library installation.

3.2 Hardware Requirements

- Processor: Intel Core i5 or higher (multi-core recommended)
- RAM: Minimum 8 GB (16 GB recommended for better performance)
- Storage: At least 256 GB (SSD preferred)
- System Type: 64-bit architecture
- Internet Connection: Required for data access and updates

CHAPTER 4

SYSTEM DESIGN

4.0 Proposed Methods

The proposed methodology outlines how each module of the AQI prediction system functions to provide accurate air quality forecasting using the GA-KELM model. The approach is designed to ensure high accuracy, efficiency, and real-time prediction capability, enabling better environmental monitoring and decision-making. The system integrates data preprocessing, machine learning, and optimization techniques to handle complex air pollution patterns.

Data Collection Module: The system begins with collecting air quality data from monitoring stations or publicly available datasets. The dataset includes pollutant concentrations such as SO_2, NO_2, PM10, PM2.5, O_3, and CO, along with meteorological factors like temperature, humidity, wind speed, and pressure. This data is stored in a structured format and serves as the primary input for the prediction model.

Data Preprocessing Module: Once the data is collected, preprocessing is performed to ensure data quality and consistency. This includes handling missing values using interpolation methods, removing outliers using statistical techniques such as the 3σ rule, and normalizing the data into a standard range. This step is essential to improve the performance and accuracy of the prediction model.

Feature Processing Module: In this stage, relevant features are selected and arranged in a time-series format to capture temporal relationships. Previous time-step values ($t-1$, $t-2$, etc.) are used as inputs to predict future AQI values. This helps the model understand patterns and trends in air pollution data.

KELM Modeling Module: The Kernel Extreme Learning Machine (KELM) is applied to model the relationship between input features and AQI values. A kernel function, typically the Radial Basis Function (RBF), is used to transform the input data into a higher-dimensional space. This enables the model to capture complex nonlinear relationships and improve prediction accuracy.

GA Optimization Module: Genetic Algorithm is used to optimize the parameters of the KELM model, including input weights, hidden layer thresholds, and kernel parameters

(C and γ). The GA starts with an initial population of solutions and evaluates them using a fitness function such as RMSE. Through selection, crossover, and mutation operations, the algorithm iteratively finds the optimal parameters, improving model performance and stability.

AQI Prediction Module: After optimization, the trained GA-KELM model is used to predict pollutant concentrations and calculate the AQI value. The system generates accurate predictions based on input data and learned patterns, making it suitable for real-time forecasting.

Output Visualization Module: The predicted AQI values and pollutant levels are presented to the user in a clear and structured format. Graphs and charts may be used to visualize trends and comparisons between actual and predicted values, helping users better understand air quality conditions.

Advantages of Methodology:

The proposed methodology provides several advantages in AQI prediction. It offers high prediction accuracy by combining kernel-based learning with genetic optimization. The system reduces instability caused by random parameter selection and improves consistency across multiple runs.

Additionally, the model ensures faster training compared to traditional neural networks, making it suitable for real-time applications. The inclusion of meteorological factors enhances prediction reliability, while preprocessing techniques improve data quality.

The system is scalable and can be extended to integrate real-time data sources and advanced visualization tools. Overall, the methodology provides an efficient, reliable, and intelligent solution for air quality prediction and environmental monitoring.

4.1 Class Diagram

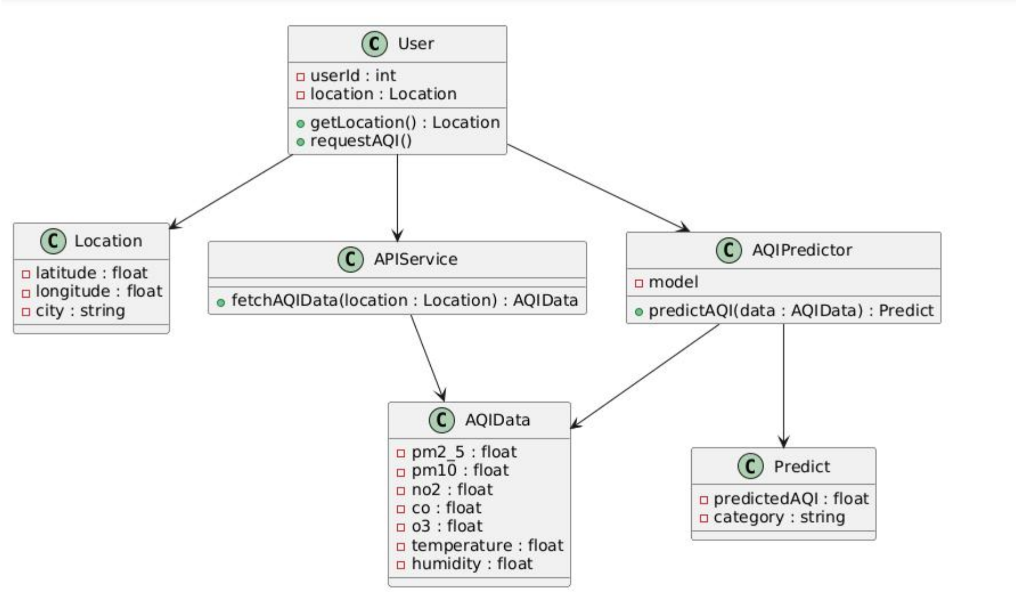

Fig 4.1.1 Class Diagram

The class diagram represents an **AQI Prediction System** showing how components interact to provide air quality predictions.

The **User** class stores user details and requests AQI using location data from the **Location** class. The **APIService** fetches real-time air quality data based on this location. This data is stored in the **AQIData** class, which includes pollutant and weather parameters.

The **AQIPredictor** processes this data using a model to predict AQI values. The result is stored in the **Predict** class, which contains the predicted AQI and its category.

Overall, the diagram shows a clear flow from user input → data collection → prediction, ensuring a simple and modular system design.

4.2 Use Case Diagram

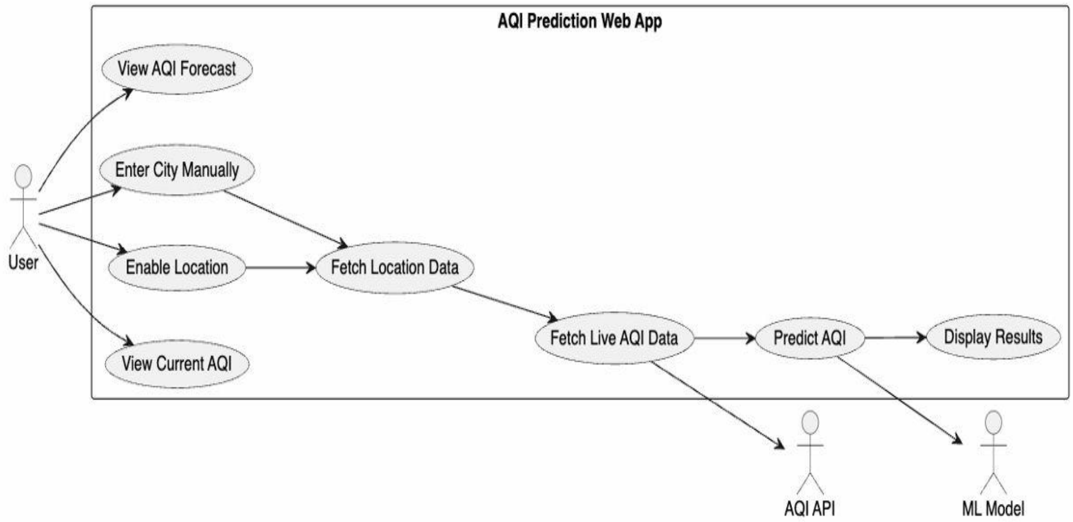

Fig 4.2.1 Use Case diagram

The use case diagram represents the interaction between the user and the AQI Prediction Web Application. The **User** is the primary actor who can perform actions such as viewing current AQI, forecasting AQI, entering a city manually, or enabling location access. Based on the user's input, the system fetches location data and retrieves live AQI data from an external **AQI API**.

The system then processes this data using the **ML Model** to predict AQI values. Finally, the predicted results are displayed to the user. The diagram shows how the system integrates user input, external data sources, and machine learning to provide accurate and real-time AQI predictions.

4.3 Activity Diagram

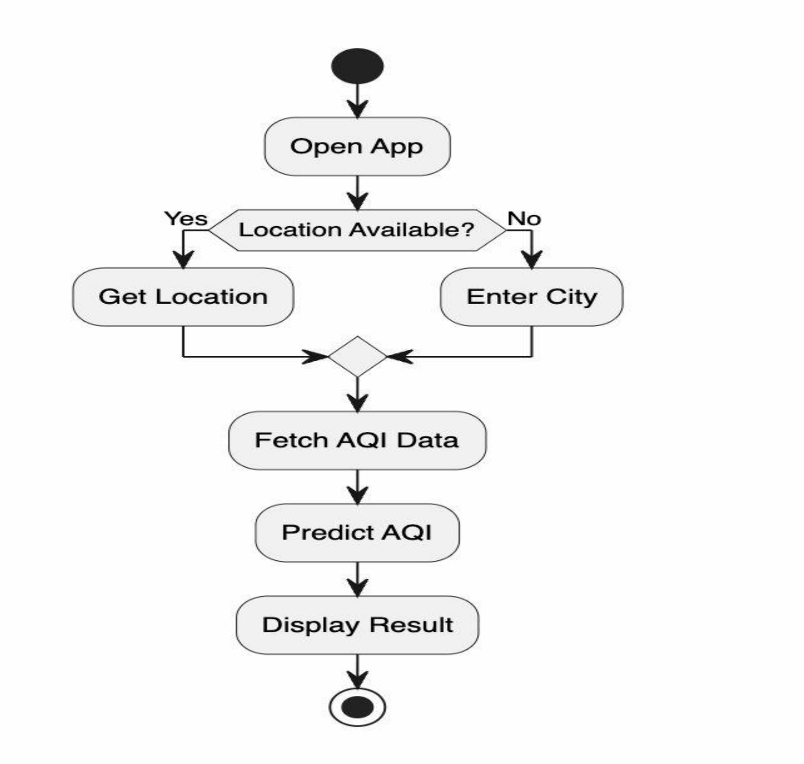

Fig 4.3.1 Activity Diagram

The activity diagram illustrates the workflow of the AQI prediction system. The process begins when the user opens the application. The system then checks whether the user's location is available. If the location is available, it is automatically fetched; otherwise, the user is prompted to enter the city manually.

After obtaining the location, the system fetches AQI data from external sources. This data is then processed by the model to predict AQI values. Finally, the predicted results are displayed to the user. The diagram clearly shows the step-by-step flow of actions and decision-making within the system.

4.4 Sequence Diagram

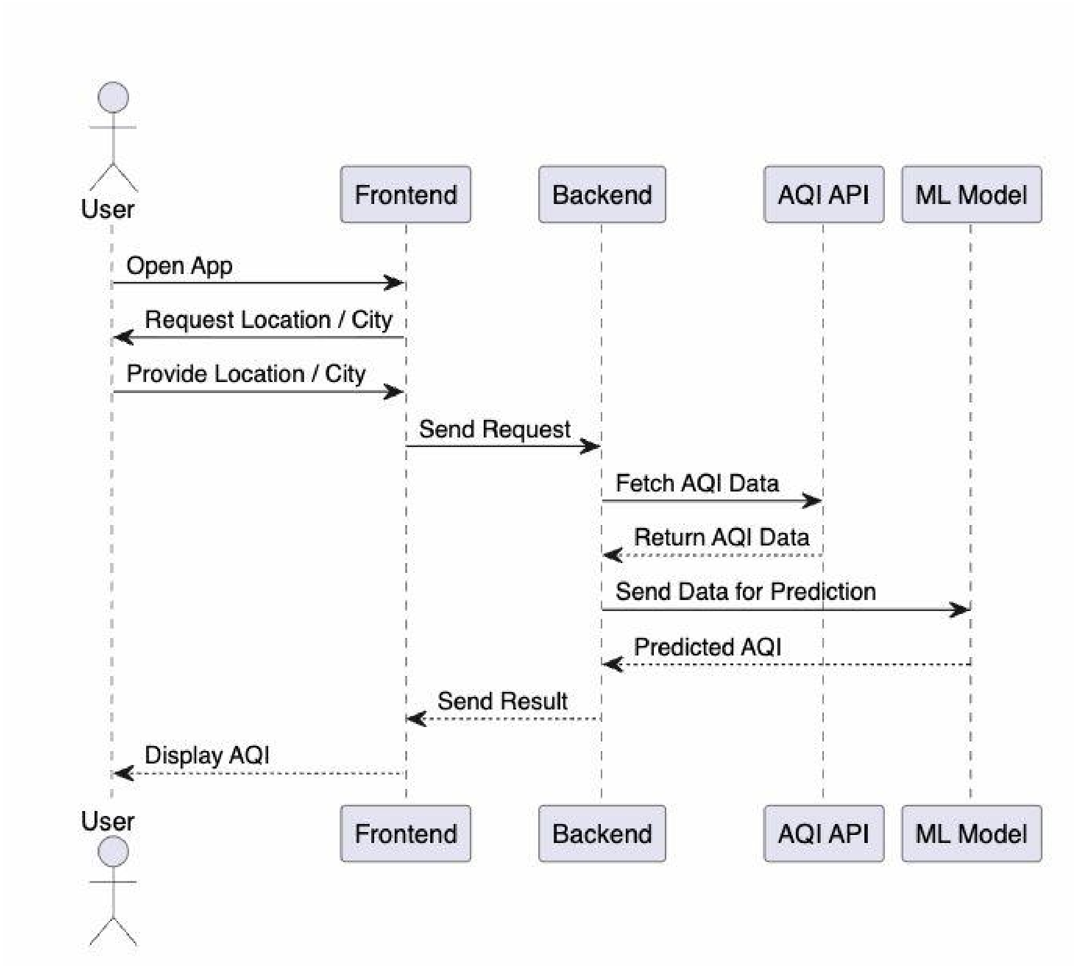

Fig 4.4.1 Sequence Diagram

The sequence diagram shows the interaction between different components of the AQI prediction system over time. The process starts when the **User** opens the application through the frontend. The frontend requests the user to provide location details, either automatically or manually.

Once the location is provided, the frontend sends a request to the backend. The backend then fetches real-time AQI data from the **AQI API** and sends this data to the **ML Model** for prediction. The ML model processes the data and returns the predicted AQI values to the backend.

Finally, the backend sends the results to the frontend, and the AQI is displayed to the user. The diagram clearly illustrates the step-by-step communication between user, frontend, backend, external API, and machine learning model.

4.5 System Architecture

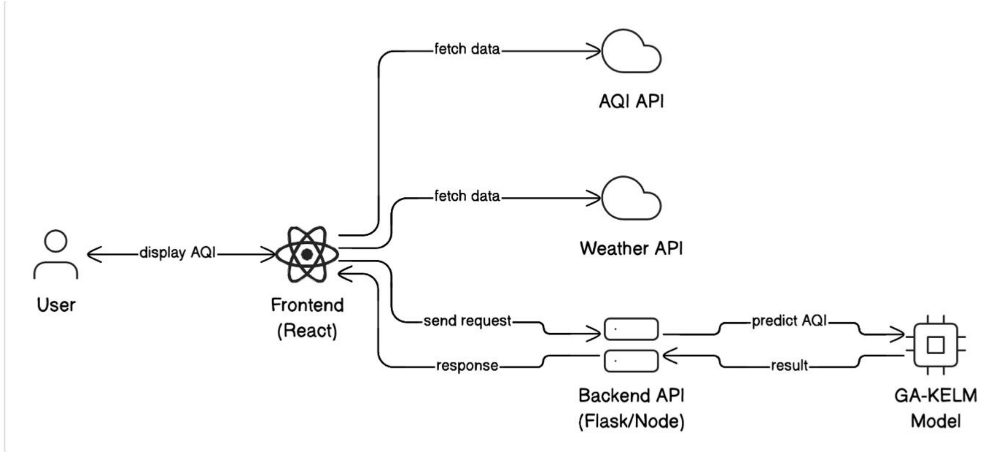

Fig 4.5.1 System Architecture

The system architecture diagram illustrates the overall working of the AQI prediction system and the interaction between its main components. The process begins with the **User**, who interacts with the system through the **Frontend (Web App)** to request AQI information.

The frontend sends the user's request to the **Backend API**, which acts as the central processing unit of the system. The backend fetches real-time AQI data from the external **AQI API** and also sends relevant data to the **AQI Prediction Model** for processing. The machine learning model analyzes the data and generates predicted AQI values.

Finally, the backend receives the prediction results and sends them back to the frontend, where they are displayed to the user. This architecture ensures smooth data flow, efficient processing, and real-time AQI prediction.

4.6 Technology Description

The **AQI Prediction System** is developed using a modern full-stack architecture that combines a responsive frontend, a powerful backend, and a machine learning model to provide accurate and real-time air quality predictions. The system is designed to ensure smooth data flow, efficient processing, and an interactive user experience.

The **frontend** is built using **React.js**, which enables the creation of dynamic and reusable user interface components. It provides a user-friendly interface where users can enter their location or allow automatic detection to view AQI results. React's virtual DOM ensures faster rendering, improving performance and responsiveness.

The **backend** is implemented using **FastAPI**, a high-performance Python framework for building APIs. It handles requests from the frontend, processes input data, and coordinates communication between external APIs and the machine learning model. FastAPI's asynchronous capabilities and automatic documentation make it efficient and developer-friendly.

The system integrates with external **AQI and Weather APIs** to fetch real-time environmental data. These APIs provide important parameters such as pollutant levels (PM2.5, PM10, NO_2, CO, O_3) and weather conditions like temperature and humidity, which are essential for accurate AQI prediction.

The core component is the **GA-KELM machine learning model**, developed using Python libraries like NumPy, Pandas, and Scikit-learn. This model captures complex relationships between pollutants and environmental factors, while the Genetic Algorithm optimizes parameters to improve prediction accuracy and stability.

Overall, the system follows a clear workflow: the frontend sends a request to the backend, which fetches real-time data, processes it using the ML model, and returns the predicted AQI along with its category. This integration ensures a scalable, efficient, and reliable AQI prediction system.

CHAPTER 5

IMPLEMENTATION & TESTING

5.1 Front page Screenshot

The front page of the AQI Prediction System serves as the primary interface for user interaction. It is designed to be simple, intuitive, and user-friendly, allowing users to easily access air quality information and predictions.

The page includes a location input section where users can either enter a city manually or use the “Use Location” option to automatically detect their current location. A search button is provided to fetch AQI data and initiate the prediction process. The interface also contains a navigation bar with options such as Home, Dashboard, and Precautions for easy navigation.

The system displays current AQI values along with their category (e.g., Moderate, Poor) using clear visual indicators. Additional sections such as safety guidelines, pollution reduction tips, and AQI maps are included to enhance user understanding. The layout is designed using HTML, CSS, and JavaScript to ensure responsiveness and better user experience.

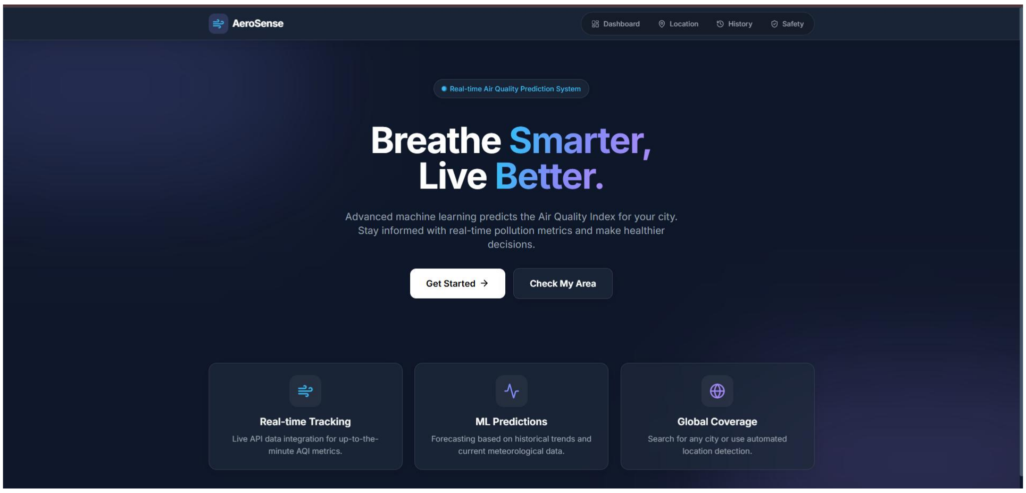

Fig 5.1.1 Front page

5.2 Results and Discussions

The AQI Prediction System was successfully implemented and tested to evaluate its performance in predicting air quality levels. The system accepts user input in the form of location data and retrieves real-time AQI information from external APIs. This data is then processed by the GA-KELM model to generate accurate AQI predictions.

The results show that the system effectively predicts AQI values and displays them in a clear and structured format. The integration of real-time data and machine learning ensures that the predictions are both accurate and relevant. The system was tested with different locations, and in most cases, it provided consistent and reliable results.

The preprocessing and model optimization techniques play a crucial role in improving prediction accuracy. The use of Genetic Algorithm ensures optimal parameter selection, while KELM handles nonlinear relationships efficiently. From a usability perspective, the system provides fast response times and an interactive interface, making it suitable for real-time applications.

However, some limitations were observed. The system's accuracy depends on the availability and quality of external API data. Additionally, extreme weather conditions or sudden pollution spikes may slightly affect prediction accuracy. Future improvements can include integrating more datasets and advanced models to enhance performance.

Overall, the results indicate that the AQI Prediction System is effective, reliable, and suitable for real-time air quality monitoring

Fig 5.2.1Dashboard

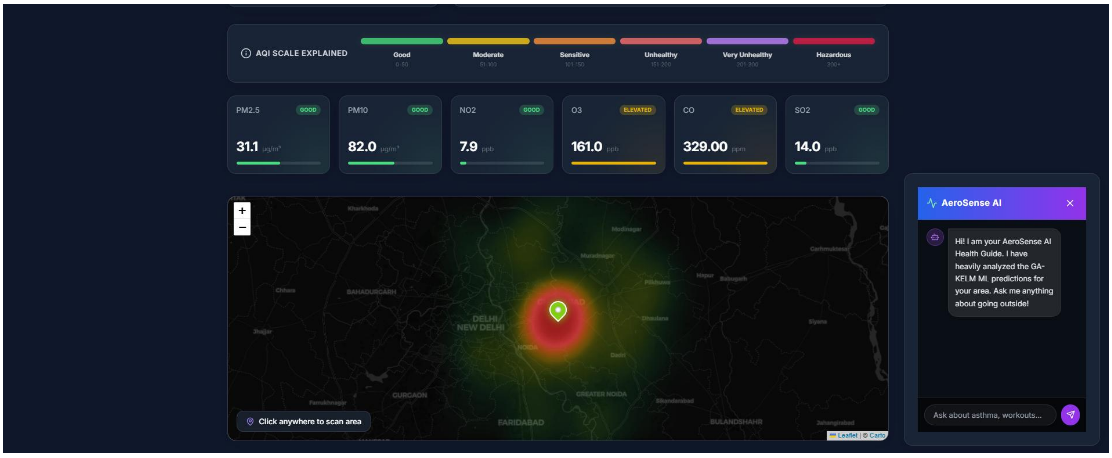

Fig 5.2.2 Map & Chatbot

5.3 Testing

Test Cases for Data Collection Module

This module was tested by providing valid latitude and longitude inputs to fetch real-time air quality data from the Open-Meteo API. Invalid coordinates and API failure scenarios were tested to verify error handling. Network delays and missing API responses were also checked to ensure system stability.

Test Cases for Data Preprocessing Module

The preprocessing module was tested using datasets containing missing values, outliers, and inconsistent data formats. Normalization and scaling were verified to ensure correct transformation. The system was tested to ensure that corrupted or incomplete data does not crash the model.

Test Cases for GA-KELM Model

The machine learning model was tested using training and testing datasets. The system was evaluated for prediction accuracy using RMSE and R^2 metrics. Different pollutant combinations were used to verify model generalization. Edge cases such as extremely high pollution values were tested to ensure stable predictions.

Test Cases for API Integration (Frontend ↔ Backend)

The communication between React frontend and FastAPI backend was tested. API endpoints such as /api/aqi and /api/forecast were verified. JSON responses were checked for correctness. Error scenarios like backend failure and invalid responses were tested to ensure proper UI handling.

Test Cases for AQI Prediction Module

The prediction module was tested with real-time pollutant data. The system correctly calculated AQI values and categorized them into levels such as Good, Moderate, and Unhealthy. Fallback logic was tested when ML model fails.

Test Cases for Forecast Module

The 7-day AQI forecast module was tested using real-time hourly data. The system correctly generated future predictions based on current data. Edge cases such as insufficient data and invalid timestamps were tested.

Test Cases for User Interface (UI Module)

The frontend UI was tested for responsiveness and usability. Features such as map interaction, location input, and AQI display were verified. Different screen sizes were tested to ensure responsiveness.

Test Cases for Overall System Integration

The complete system flow from user input → API → ML model → output display was tested. Performance and response time were measured. Multiple user requests were tested to ensure system stability and scalability.

Test ID	Module	Test Case	Expected Result	Status
UT-01	Data Collection	Valid lat/lng input	AQI data fetched successfully	Pass
UT-02	Data Collection	Invalid coordinates	Error handled properly	Pass
UT-03	Preprocessing	Missing values in dataset	Data cleaned and normalized	Pass
ML-01	GA-KELM Model	Training with dataset	Model trained successfully	Pass
ML-02	GA-KELM Model	Prediction with test data	Accurate AQI predicted	Pass
IT-01	Frontend → Backend	User selects location	API request sent successfully	Pass
IT-02	Backend → Model	Pollutant data sent to model	AQI predicted correctly	Pass
IT-03	Backend API	/api/aqi endpoint	JSON response returned	Pass
IT-04	Forecast Module	7-day forecast request	Forecast data displayed	Pass
UT-04	AQI Module	High pollution input	AQI categorized correctly	Pass
UT-05	UI Module	Map click interaction	Location selected correctly	Pass
UT-06	UI Module	Invalid user input	Error message shown	Pass
ST-01	System Testing	End-to-end flow	Full system works correctly	Pass

CHAPTER 6

CONCLUSION

6.1 Conclusion

The AQI Prediction System using the GA-KELM model has been successfully developed and implemented to provide accurate and efficient air quality forecasting. The system effectively combines Kernel Extreme Learning Machine with Genetic Algorithm optimization to overcome the limitations of traditional machine learning models. By handling nonlinear relationships and automatically optimizing parameters, the model achieves better prediction accuracy and stability.

The system was tested using real-world air quality data, and the results demonstrate that it provides reliable and consistent AQI predictions. The integration of preprocessing techniques and meteorological factors further enhances the model's performance. Additionally, the system offers a user-friendly interface and fast response time, making it suitable for real-time applications.

Overall, the project provides an effective solution for air quality monitoring and prediction. It can assist individuals and authorities in taking preventive measures to reduce the impact of air pollution. The GA-KELM approach proves to be a powerful and efficient method for solving complex environmental prediction problems.

6.2 Future Scope

The AQI Prediction System can be further enhanced in several ways to improve its performance and usability. One possible improvement is the integration of real-time data from IoT-based air quality sensors, which would provide more accurate and location-specific predictions. This would make the system more reliable for real-world applications.

Another area of enhancement is the development of a mobile application that allows users to access AQI predictions on the go. Features such as alerts and notifications for high pollution levels can help users take timely precautions. Integration with government and environmental monitoring systems can also make the application more impactful.

Furthermore, the system can be extended to provide personalized health recommendations based on AQI levels, such as suggesting safe outdoor activities or precautions. Overall, these improvements can make the system more intelligent, scalable, and useful for environmental monitoring and public health awareness.

REFERENCES

[1] C. Liu, G. Pan, D. Song and H. Wei, "Air Quality Index Forecasting via Genetic Algorithm-Based Improved Extreme Learning Machine," IEEE Access, vol. 11, pp. 67086–67095, 2023.

[2] G.-B. Huang, Q.-Y. Zhu and C.-K. Siew, "Extreme Learning Machine: Theory and Applications," Neurocomputing, vol. 70, no. 1–3, pp. 489–501, 2006.

[3] V. Vapnik, The Nature of Statistical Learning Theory, New York, NY, USA: Springer, 1995.

[4] J. H. Holland, "Genetic Algorithms," Scientific American, vol. 267, no. 1, pp. 66–73, 1992.

[5] S. Haykin, Neural Networks and Learning Machines, 3rd ed., Pearson, 2009.

[6] X. Xue et al., "Air Quality Prediction Based on Genetic Algorithm Optimized Extreme Learning Machine," Environmental Science and Pollution Research, 2022.

[8] Chunhao Liu, "AQI Dataset," GitHub Repository, [Online]. Available:

[9] World Health Organization (WHO), "Air Pollution and Health," [Online]. Available:

[10] J. Zheng et al., "Forecasting Fine-Grained Air Quality Based on Big Data," Proceedings of the 21th ACM SIGKDD International Conference, 2015.